Density amplification in a largest-prime-factor dynamical system

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Abstract

For a finite set of primes S, retain its elements and adjoin $P^+(2(p+q)-1)$ for every $p, q \in S$, allowing $p = q$ at every generation. We prove that there are constants $\eta > 0$ and M_0 such that any finite seed missing at most $\eta M/\log M$ primes up to one scale $M \geq M_0$ generates a set containing all but $O_{S,M}(X^{79/80})$ primes up to X. A stronger statement controls the actual finite generations at geometrically increasing cutoffs. The proof couples a power-saving exceptional-set estimate for canonical prime-pair representations with a deletion estimate uniform over all missing input sets. Exact representation-degree normalization and an averaged singular-factor overlap bound give a defect multiplier $17/16$, smaller than the available scale growth. We give the complete iteration and interpolation argument, identify the analytic inputs and their normalizations, and record the corresponding Lean declarations. The motivating conjectures assert that the seed $\{2\}$ generates every prime, and that each odd-prime singleton generates every odd prime. The theorem applies to sufficiently rich finite seeds; neither conjecture is proved here.

1 Introduction

Largest-prime-factor dynamics combines two operations with rather different behavior. Addition produces many values from a finite set; taking the largest prime factor merges many of those values into a single output. A large number of pairs therefore need not give a large number of distinct new primes. This paper studies an accumulating, branching system in which that collision problem can be controlled after a suitable finite density condition has been reached.

Write $P^+(n)$ for the largest prime factor of $n > 1$. For a finite set S of primes define

$$\mathcal{T}(S) = S \cup \{P^+(2(p+q)-1) : p, q \in S\}, \qquad S_0 = S, \quad S_{j+1} = \mathcal{T}(S_j). \tag{1}$$

Pairs are ordered when counted, although the set of outputs is unchanged by interchanging the two inputs. Diagonal pairs are always permitted. The generated closure is $\mathcal{C}(S) = \bigcup_{j \geq 0} S_j$.

Our main result is a robust propagation theorem: a sufficiently dense finite prime prefix amplifies into density one, with a quantitative power-saving exceptional set. The robustness is important. We impose no independence assumption, no random model, and no residue-distribution hypothesis on the missing primes of the seed. Their effects are controlled jointly through an incidence matrix.

1.1 The motivating generation conjectures

Let $\mathcal{P}$ denote the set of all primes. The original objective is complete prime generation, not merely relative density one.

Conjecture 1.1 (Generation from two)**.** For the dynamics (1),

$$\mathcal{C}(\{2\}) = \mathcal{P}.$$

Equivalently, for every prime r there is a finite generation j with $r \in S_j$ when $S_0 = \{2\}$.

Conjecture 1.2 (Generation from an odd prime)**.** For every odd prime p,

$$\mathcal{C}(\{p\}) = \mathcal{P} \setminus \{2\}.$$

The generation at which a target prime appears may depend on both the initial prime p and the target. Repeated inputs remain permitted at every generation.

The missing prime 2 in Conjecture 1.2 is forced: every new output is odd. The odd-singleton conjecture also implies Conjecture 1.1. Indeed $P^+(2(2+2)-1) = 7$, and monotonicity of the closure gives $\mathcal{C}(\{7\}) \subseteq \mathcal{C}(\{2\})$. If Conjecture 1.2 holds even just for $p = 7$, this inclusion supplies all odd primes, while 2 is present initially.

The density-amplification theorem below is a step toward these generation questions, rather than their replacement. Even proving dense entry from a singleton would leave a further gap: the resulting $O(X^{79/80})$ exceptional set may contain infinitely many primes. Complete generation requires showing that each individual prime is eventually reached, including any primes left outside that density conclusion.

1.2 Related work and the scope of the contribution

Caragiu and Vicol studied prime magmas with operation $P^+(ap+bq)$, including accumulation from all available pairs [2]. Their cyclicity questions and finite-generation examples are close in spirit, but their homogeneous operations are different from the affine operation in (1). Back and Caragiu considered greatest-prime-factor recurrences with two preceding inputs [1]; these are single trajectories, rather than branching closures. Neither comparison identifies an existing theorem about the specific closure studied here.

The analytic setting also has substantial antecedents. Ramaré and Ruzsa developed additive results for dense subsets of sifted sequences using sieve methods [3]. Restriction estimates for sieve majorants provide another approach to additive questions in the primes [4]. Alsetri and Shao showed that even prime subsets with arbitrarily high global relative density can miss a positive proportion of even sums; they also established an almost-all binary Goldbach theorem under uniform relative-density conditions in reduced residue classes [5]. These results explain why a density-to-representation passage requires care. In particular, an unspecified $o(X)$ exceptional set among integers is insufficient when the desired targets are indexed by only $O(X/\log X)$ primes.

The contribution here is the combination of canonical largest-prime ownership, a deletion estimate uniform over the actual missing input set, and iteration on the *generated states*. The exponent 79/80 is inherited from a deliberately conservative analytic supply estimate; it is not asserted to be optimal. The character estimates used by the installed proof chain draw on the 2026 zero-free work of OpenAI [6]. We distinguish that analytic source from the dynamical argument and state the exact estimate consumed below. We make no claim to a new zero-free region, a new Goldbach exponent, or a resolution of the small-seed reachability problem.

2 An informal explanation of the argument

Imagine that the current set already contains almost every prime below a large number M. We try to produce almost every prime in the next window, from M to roughly $5M/4$. A

target prime r will be produced if we can find two available primes whose sum is a carefully chosen even number.

There are two choices for that even number. If r is 3 modulo 4, use $(r+1)/2$. Then the value before taking its largest prime factor is exactly r. If r is 1 modulo 4, use $(3r+1)/2$. The value is then $3r$, whose largest prime factor is again r. Thus the multiplicative output question becomes a controlled pair of additive representation questions.

Almost every target has many representations using the full prime prefix. Some of those representations can disappear because an input prime is missing from the current state. A crude argument charging each missing input separately loses too much. Instead we examine all failed targets together. If a target fails, every representing pair must touch a missing input. Cauchy–Schwarz converts that statement into an energy inequality. The diagonal part of this energy is known exactly after dividing each target row by its own representation count. The overlap between two different rows is bounded by a three-prime sieve and then averaged over the actual failed targets.

This gives the key comparison. The number of holes at the next cutoff is at most $17/16$ times the old number, plus a power-saving exceptional term and a logarithmic remainder. Meanwhile the cutoff grows by roughly $5/4$. The holes therefore occupy a smaller and smaller fraction of the prime prefix. Once this vanishing fraction is available, the stronger power-saving recurrence can be used permanently. Its forcing term has size $M^{79/80}$; geometric summation gives the same exponent for the holes themselves.

Finally, a prime that is missing forever is missing at every finite generation. Choosing the first cutoff above an arbitrary X transfers the finite-generation bound to the permanently missing primes below X. This last step matters: a bound only at a sparse sequence of cutoffs is not yet the advertised statement for every large X.

The argument begins *after dense entry*. The fact that a small seed has many pairs, or behaves well modulo a few small primes, does not prove that it reaches this regime. The seed $\{2\}$ remains the main unresolved instance for this method. Our ultimate target is the complete generation assertion in Conjecture 1.1, together with its odd-singleton counterpart, Conjecture 1.2.

3 Definitions and main results

Let

$$\mathcal{P}(M) = \{p \le M : p \text{ is prime}\}, \qquad W(M) = \mathcal{P}(\lfloor 5M/4 \rfloor) \setminus \mathcal{P}(M), \qquad N(M) = \frac{M}{\log M}.$$

All asymptotic statements concern positive integer M tending to infinity. Floors in cutoffs are retained. Define

$$m_0 = M, \qquad m_{j+1} = \lfloor 5m_j/4 \rfloor, \qquad E_j = |\mathcal{P}(m_j) \setminus S_j|, \qquad \gamma = \frac{79}{80}. \tag{2}$$

The iterated cutoffs m_j are distinct from the fixed entry threshold.

Theorem 3.1 (Power-saving propagation)**.** *There exist $\eta > 0$ and an integer $M_{\text{entry}} \ge 8$ such that, for every $M \ge M_{\text{entry}}$ and every finite set S of primes satisfying*

$$|\mathcal{P}(M) \setminus S| \le \eta \frac{M}{\log M}, \tag{3}$$

there are $K > 0$ and an integer J for which

$$|\mathcal{P}(m_j) \setminus S_j| \le K m_j^{79/80} \qquad (j \ge J). \tag{4}$$

In particular, for some $K' > 0$ and X_0,

$$|\{r \le X : r \text{ prime},\ r \notin \mathcal{C}(S)\}| \le K' X^{79/80} \qquad (X \ge X_0). \tag{5}$$

The constants K, J, K', X_0 may depend on the fixed seed and entry scale. The entry fraction and entry threshold precede the quantifiers over M and S.

In fact the Lean statements allow an arbitrary finite set of natural numbers as the initial set: the defect counts only missing primes. We state the prime-seed specialization to keep the dynamical interpretation uniform.

Corollary 3.2. *For every seed satisfying Theorem 3.1,*

$$\frac{|\mathcal{C}(S) \cap \mathcal{P}(X)|}{\pi(X)} \longrightarrow 1.$$

Consequently $\mathcal{C}(S)$ is infinite. The fraction of primes below X that remain unreachable is $O_{S,M}(X^{-1/80} \log X)$.

Proof. Divide (5) by $\pi(X) \sim X/\log X$. Since $X^{-1/80} \log X \to 0$, the missing fraction tends to zero. An infinite number of primes is therefore reached. □

Corollary 3.3 (Finite-stage appearance)**.** *Under the same hypotheses, all but $O_{S,M}(X^{79/80})$ primes up to X are present after $O_M(\log X)$ generations. The lower medians of the increasing finite prime sets S_j tend to infinity.*

Proof. For $m \ge 8$, $\lfloor 5m/4 \rfloor \ge 9m/8$. The first index j with $m_j \ge X$ is therefore at most $\lceil \log(X/M)/\log(9/8) \rceil$ when $X > M$. For sufficiently large X, that index is at least J, and $m_j \le 5X/4$. Apply (4). To prove the median assertion, fix B. There are at most $\pi(B)$ primes at most B. The sets S_j are nested and their union is infinite, so their cardinalities tend to infinity. Eventually more than half their elements exceed B. □

The two corollaries above are ordinary mathematical consequences of the formalized main theorem and the prime number theorem. They are not advertised as separate Lean declarations in this release.

4 Canonical prime targets and exact deletion

For an odd prime r, put

$$c_r = \begin{cases} 1 & r \equiv 3 \pmod 4, \\ 3 & r \equiv 1 \pmod 4, \end{cases} \qquad t_r = \frac{c_r r + 1}{2}. \tag{6}$$

Then t_r is even and $2t_r - 1 = c_r r$. Since every prime factor of c_r is at most r, we have $P^+(c_r r) = r$. Therefore

$$p + q = t_r, \quad p, q \in S \quad \Longrightarrow \quad r \in \mathcal{T}(S). \tag{7}$$

This includes $p = q$.

For a finite prime set P and an even integer t, define the *ordered* representation degree

$$R_P(t) = |\{p \in P : t - p \in P\}|,$$

where subtraction is interpreted only when $p \le t$. Every representation is counted by its first input; a diagonal representation contributes once. Write $R_P(r) = R_P(t_r)$ when the

target prime is clear. For $D = P \setminus S$, the absence of a representation with both inputs in S implies

$$R_P(r) \le 2|\{p \in D : t_r - p \in P\}|. \tag{8}$$

Indeed the involution $p \mapsto t_r - p$ pairs the representations; every pair meets D. The same inequality is valid at a fixed point of the involution, so no diagonal correction is discarded.

Lemma 4.1 (Distinct targets)**.** *For $M \ge 3$, the map $r \mapsto t_r$ is injective on $W(M)$.*

Proof. Within either residue branch this is immediate. An equality across branches would give $r = 3s$ or $s = 3r$, whereas both targets lie in $(M, 5M/4]$. Their ratio is less than $5/4$, so this is impossible. □

We use the finite divisor factor

$$\sigma(n) = \prod_{\substack{\ell \mid n \\ \ell \text{ prime}, \ell \ge 3}} \left(1 + \frac{1}{\ell - 2}\right), \qquad \sigma(1) = 1. \tag{9}$$

Only nonzero differences occur when $\sigma(|t_r - t_s|)$ is used. The two normalizations in the proof have different roles: $\sigma(t_r)^{-1}$ removes the arithmetic factor from a supply or sieve bound, while $R_P(r)^{-1}$ makes each target row have mass exactly one.

5 The analytic representation supply

We state the quantitative inputs with their ranges and normalizations. They are proved in the maintained Lean source, rather than introduced as new assumptions of Theorem 3.1.

5.1 Character input and the complete error

The centered character estimate consumed by the major-arc calculation is

$$\left|\sum_{n \le x} \Lambda(n)\chi(n) - x\,\mathbf{1}_{\chi=\chi_0}\right| \le A x^{15/16}(1 + \log q + \log(x+2))^2 \qquad (x \ge 65536). \tag{10}$$

Here one constant $A > 0$ precedes every modulus $q \ge 1$, character χ mod q, and prefix x. The principal character includes the finite Euler-factor correction in the proof. The nonprincipal branch uses the installed $7/8$ character estimate, while the principal branch is relaxed to $15/16$. The formal owner is `QRH.exists_all_characterChebyshevSum_power_bound`. The zero-free theorem in [6] is part of the analytic provenance; we use the quantitative estimate (10), not a bare zero-free statement with an unstated uniformity inference.

For an integer interval $I = (s, s + L]$, let

$$V_I(t) = \sum_{\substack{a,b \in I \\ a+b=t}} \Lambda(a)\Lambda(b), \qquad H_I(t) = |\{(a,b) \in I^2 : a + b = t\}|.$$

Use the standard binary Goldbach singular series

$$\mathfrak{S}(t) = \begin{cases} 2\displaystyle\prod_{\ell > 2}\left(1 - \frac{1}{(\ell-1)^2}\right)\sigma(t), & t \text{ even}, \\ 0, & t \text{ odd}. \end{cases}$$

The error is centered at the *infinite* series: $\mathcal{E}_I(t) = V_I(t) - H_I(t)\mathfrak{S}(t)$.

Lemma 5.1 (Complete fixed-interval error)**.** *There is $A_1 > 0$ such that, for every integer $M \geq 2^{18}$, every interval with*

$$M/4 \leq s, \quad s + L \leq 2M, \quad L \leq M,$$

and every $T \subseteq \{1, \dots, 2M\}$,

$$\sum_{t \in T} |\mathcal{E}_I(t)|^2 \leq A_1 M^{239/80}. \tag{11}$$

The constant is uniform in the interval and target set.

Here is the error accounting behind this bound. Choose an integer P with $\frac{1}{2}M^{1/40} \leq P \leq M^{1/40}$. The cyclic grid in the formal proof has a period $N > 2M$, divisible by M and every $q \leq P$; one may enlarge such a period to ensure $s + L < N$. Consequently neither the interval nor the target set wraps around the grid. Inverse Fourier inversion gives the exact identity

$$\mathcal{E}_I(t) = \mathcal{E}_{\mathrm{maj}}(t) + \mathcal{E}_{\min}(t) + \mathcal{E}_{\ker}(t) + H_I(t)(\mathfrak{S}_P(t) - \mathfrak{S}(t)). \tag{12}$$

The kernel packet retains the interval weights; the last packet retains the actual pair count. Neither is replaced by an unweighted tail.

For completeness, the squared packet budgets used in the proof, with $L_M = 1 + \log(2M)$, are bounded by constants times

$$(8K^2(1 + \log M)^4 + 6K^4(1 + \log M)^8)M^{119/40}, \qquad \text{major}, \tag{13}$$

$$CM^{1/160}(M^3/P + M^{13/5})L_M^9, \qquad \text{minor}, \tag{14}$$

$$BM^{481/160}/P^2, \qquad \text{kernel}, \tag{15}$$

$$DM^{481/160}/P, \qquad \text{series tail}. \tag{16}$$

All four constants are positive and independent of M, N, I, T. The major bound uses (10) after the character expansion at reduced rational frequencies. The minor bound uses the installed Vaughan decomposition and its mean-square estimate. The last two bounds use Fourier inversion for the interval kernel and divisor moment bounds for the singular tail. The exact finite Fourier identities and the four estimates have separate formal owners listed in Appendix A; thus an overlap or kernel error is not hidden inside an unspecified error term.

Proof of Lemma 5.1 from the packet bounds. Apply $|u_1 + \cdots + u_4|^2 \leq 4\sum_i |u_i|^2$ to (12) and sum over the same target set. Substituting $P \asymp M^{1/40}$ in (13)–(16) bounds the result by

$$A_2 M^{477/160}(1 + \log(2M))^9.$$

For example, the minor term $M^{1/160}M^3/P$ has exponent $477/160$, the kernel term has exponent $473/160$, and the series tail again has exponent $477/160$. The other terms are smaller. The elementary uniform bound

$$(1 + \log(2M))^9 \leq 1442^9 M^{1/160} \qquad (M \geq 1)$$

absorbs the remaining logarithms. Since $478/160 = 239/80$, (11) follows. The formal proof constructs P by an integer cutoff and includes the factor-two rounding bounds. □

The packet estimates are analytic library results, with their own proofs in the companion formal development. The elementary argument above records their complete consumer; it does not purport to reprove the underlying character and minor-arc theorems from first principles.

5.2 Supply on the two canonical branches

Proposition 5.2 (Positive supply)**.** *There is $C_{\mathrm{src}} > 0$ such that, for every $M \geq 2^{44}$, there is $Z_M \subseteq W(M)$ with*

$$|Z_M| \leq C_{\mathrm{src}} M^{79/80}$$

and, for every $r \in W(M) \setminus Z_M$,

$$\frac{R_{\mathcal{P}(M)}(r)}{\sigma(t_r)} \geq a \frac{M}{\log^2 M}, \qquad a = \frac{63}{1024}. \tag{17}$$

Proof. For the branch $r \equiv 1 \pmod 4$, use the fixed input interval $(\lceil M/4 \rceil, M]$. For $r \equiv 3 \pmod 4$, use the analogous interval at scale $\lfloor M/2 \rfloor$. Both inputs are at most M. The interval-pair geometry, with all integer endpoints retained, gives the positive comparison underlying

$$V_I(t_r) \geq \frac{1}{16} M \sigma(t_r)$$

outside an exceptional set. To see the size of that set, a failure of this lower bound gives a fixed positive multiple of $M\sigma(t_r)$ in the error $\mathcal{E}_I(t_r)$. Since $\sigma(t_r) \geq 1$, (11) and Chebyshev's inequality bound the number of failures by

$$O(M^{239/80}/M^2) = O(M^{79/80}).$$

The map in Lemma 4.1 prevents a multiplicity loss when passing from even targets to primes. The two interval branches are counted together in the constant.

The weights $\Lambda(a)\Lambda(b)$ include prime powers. Their total unwanted contribution is at most $4\sqrt{M}\log^2 M$; the explicit comparison at $M \geq 2^{44}$ bounds this by $M/1024$. After discarding prime powers, each remaining weight is at most $\log^2 M$. Using $\sigma(t_r) \geq 1$, the remaining weighted supply is at least

$$\left(\frac{1}{16} - \frac{1}{1024}\right) M\sigma(t_r) = \frac{63}{1024} M\sigma(t_r).$$

Division by $\log^2 M$ proves (17). The finite interval geometry and exceptional-set comparison are formalized in `LPFWideSource`; the prime-power subtraction is performed before the conversion to ordered prime representations. □

The endpoint 2^{44} is a source-conversion cutoff. It is not the entry threshold of Theorem 3.1.

6 Uniform deletion control

Put $P = \mathcal{P}(M)$, $D = P \setminus S$, $E = |D|$, and $G = W(M) \setminus Z_M$. For $r \in G$, let

$$w_r(p) = \frac{\mathbf{1}_{\{p \in P,\ t_r - p \in P\}}}{R_P(r)}.$$

By (17), every denominator is positive. The row sums are exactly one, and their diagonal energies satisfy

$$\sum_{p \in P} w_r(p) = 1, \qquad \sum_{p \in P} w_r(p)^2 = \frac{1}{R_P(r)} \leq U, \qquad U = \frac{\log^2 M}{aM}. \tag{18}$$

The three-prime upper sieve and exact normalization give, for distinct $r, s \in G$,

$$\sum_{p \in P} w_r(p) w_s(p) \le V \sigma(|t_r - t_s|), \qquad V = \frac{C_3(M/\log^3 M)}{(aM/\log^2 M)^2}. \tag{19}$$

The arithmetic factors $\sigma(t_r)\sigma(t_s)$ in the unnormalized sieve bound are divided out before applying the source lower bounds. Using only a pointwise bound on the separation factor would be too expensive.

Lemma 6.1 (Averaged overlap)**.** *There is $C_{\mathrm{ov}} > 0$ such that, for every sufficiently large M and every $B \subseteq W(M)$ of size $b \ge M/\log^4 M$,*

$$\sum_{r \in B} \sum_{\substack{s \in B \\ s \ne r}} \sigma(|t_r - t_s|) \le C_{\mathrm{ov}} b^2 \log\!\left(3 + \log \frac{N(M)}{b}\right). \tag{20}$$

The formal proof obtains a uniform high-moment bound for the separation factor and applies Hölder with $q = \lceil \log(N(M)/b) \rceil$. In the stated range, $q \ll \log\log M$, $b \le N(M)/2$ for large M, and the moment estimate costs $O(\log(q+2))$. This gives precisely the double logarithm in (20). Both residue branches and the omitted diagonal are retained. The lemma is `exists_uniform_primeTarget_overlap`; it is uniform over the subset B, which is essential for generated states.

Lemma 6.2 (Joint failed-target bound)**.** *Let $B = G \setminus \mathcal{T}(S)$, $b = |B|$, and suppose $b \ge M/\log^4 M$. If*

$$8EVC_{\mathrm{ov}} \log\!\left(3 + \log \frac{N(M)}{b}\right) \le 1, \tag{21}$$

then $b \le 8EU$.

Proof. By (8), each failed target satisfies $1 \le 2\sum_{p \in D} w_r(p)$. Sum over B and apply Cauchy–Schwarz:

$$b^2 \le 4E \sum_{p \in D} \left(\sum_{r \in B} w_r(p)\right)^2 \le 4E \sum_{p \in P} \left(\sum_{r \in B} w_r(p)\right)^2.$$

Expanding the square, (18), (19), and (20) bound the last sum by

$$bU + b^2 V C_{\mathrm{ov}} \log\!\left(3 + \log \frac{N(M)}{b}\right).$$

Move the second contribution to the left using (21). This gives $b^2/2 \le 4EbU$. Since $b > 0$, divide by $b/2$. This is a joint estimate; no distribution model for D was used. □

Proposition 6.3 (Row-normalized recurrence)**.** *For each $a > 0$, there is $\eta_a > 0$ such that, for every sufficiently large M, every finite set S, and every exceptional set Z satisfying* (17) *on $W(M) \setminus Z$,*

$$E \le \eta_a N(M) \quad \Longrightarrow \quad |\mathcal{P}(\lfloor 5M/4 \rfloor) \setminus \mathcal{T}(S)| \le \frac{17}{16} E + |Z| + \frac{8\eta_a}{a} \log M.$$

Proof. Write $F = M/\log^4 M$ and again let b count the good new targets that fail. We prove

$$b \le E/16 + (8\eta_a/a) \log M. \tag{22}$$

There is nothing to prove when $b < E/16$. Otherwise $b \geq E/16$. Put $d = E/N(M)$ and $z = N(M)/b$. If $E > 0$, then $0 < d \leq \eta_a$ and $1 \leq z \leq 16/d$. The elementary limit

$$d \log(3 + \log(16/d)) \longrightarrow 0 \qquad (d \downarrow 0)$$

allows η_a to be chosen so small that

$$8d(C_3 C_{\mathrm{ov}}) \log(3 + \log z) \leq a^2 \quad (0 < d \leq \eta_a,\ 1 \leq z \leq 16/d).$$

After substituting V, this is exactly (21). Thus, if also $b \geq F$, Lemma 6.2 gives

$$b \leq 8EU \leq (8\eta_a/a) \log M.$$

When $E = 0$, the half-deleted inequality already forces $b = 0$.

It remains to handle $0 < b < F$. Choose any failed good target. Its supply and (8), together with $\sigma(t_r) \geq 1$, imply $aM/\log^2 M \leq 2E$. Decrease η_a, if necessary, so that $64\eta_a \leq a$ and $64\eta_a \leq a^2$. Combining the supply with $E \leq \eta_a M/\log M$ gives $a \leq 2\eta_a \log M$, hence $a \log^2 M \geq 32$. Consequently

$$F = \frac{M}{\log^4 M} \leq \frac{aM}{32 \log^2 M} \leq E/16.$$

This proves (22) in the small-b case as well.

An old hole contributes at most E, an exceptional new target at most $|Z|$, and a failed good target at most b. Thus

$$|\mathcal{P}(\lfloor 5M/4 \rfloor) \setminus \mathcal{T}(S)| \leq E + |Z| + b.$$

Use (22). This retains all old holes; it does not assume they are healed. □

Corollary 6.4 (Complete one-step estimate)**.** *There exist $C, D, \eta_{\mathrm{pow}} > 0$ and a scale threshold such that, for every larger M and every finite S with $E \leq \eta_{\mathrm{pow}} N(M)$,*

$$|\mathcal{P}(\lfloor 5M/4 \rfloor) \setminus \mathcal{T}(S)| \leq \frac{17}{16} E + CM^{79/80} + D \log M. \tag{23}$$

Proof. Use Proposition 5.2 in Proposition 6.3, and take $D = 8\eta_{\mathrm{pow}}/a$. The installed library also provides a proof through its earlier logarithmic deletion recurrence; either version has exactly the form (23). □

7 Iteration: density zero and the 79/80 bound

We first record why entry leads to vanishing relative defect. This intermediate step lets us use independently selected small density constants without identifying them.

Lemma 7.1 (Vanishing defect after entry)**.** *There are $\eta_{\mathrm{ent}} > 0$ and M_{ent} such that (3) with these constants implies*

$$\frac{E_j}{m_j/\log m_j} \longrightarrow 0.$$

Proof. Since $\gamma < 1$, the source exception satisfies $CM^{\gamma} = O(M/\log^4 M)$. The logarithmic remainder is also $O(M/\log^4 M)$. Hence for some fixed $C_0 > 0$, the complete one-step inequality in its applicable density range gives

$$E_{j+1} \leq \lambda E_j + C_0 F_j, \quad \lambda = 17/16, \quad F_j = m_j/\log^4 m_j.$$

For sufficiently large starting M, both $N_j = m_j/\log m_j$ and F_j grow by at least $9/8$ at each step. Choose η_{ent} within the permitted density range and then take M_{ent} large enough that $C_0F_j \le \eta_{\mathrm{ent}}N_j/16$ throughout the increasing cutoff sequence. Induction gives $E_j \le \eta_{\mathrm{ent}}N_j$, so every application of the recurrence is justified.

Iteration, using $F_{j+1} \ge (9/8)F_j$, gives

$$E_j \le \lambda^j E_0 + 16C_0F_j.$$

For example, the inductive step is valid because $16C_0(9/8 - \lambda) = C_0$. Divide by N_j. The first term is bounded by $(17/18)^j E_0/N_0$, and the second is $16C_0/\log^3 m_j$. Both tend to zero. Floors cause no difficulty because all the scale comparisons hold beyond a single fixed threshold. □

Lemma 7.2 (Power growth)**.** *For $M \ge 8$, the cutoffs* (2) *satisfy*

$$m_j \ge M, \qquad \left\lfloor \frac{5m_j}{4} \right\rfloor \ge \frac{9}{8}m_j, \qquad m_{j+1}^{\gamma} \ge \frac{13}{12}m_j^{\gamma}.$$

Proof. The first assertion follows inductively. For an integer $m \ge 8$, the floor error in $5m/4$ is at most $3/4$, and the stated $9/8$ lower bound follows. Finally,

$$\left(\frac{13}{12}\right)^4 < \left(\frac{9}{8}\right)^3, \qquad \gamma = \frac{79}{80} \ge \frac{3}{4}.$$

Taking fourth roots and then using monotonicity in the exponent gives $(9/8)^{\gamma} \ge 13/12$. Raise the cutoff comparison to the positive power γ. □

Lemma 7.3 (Eventual recurrence with a growing forcing scale)**.** *Suppose $F_j > 0$, $F_{j+1} \ge (13/12)F_j$ for every j, and, for all $j \ge J$,*

$$E_{j+1} \le (17/16)E_j + C_1F_j.$$

Then there is $K > 0$ such that $E_j \le KF_j$ for every $j \ge J$. One may take

$$K = \max\{1, E_J/F_J, 48C_1\}.$$

Proof. At $j = J$ this is the definition of K. Since $13/12 - 17/16 = 1/48$, we have $C_1 \le K/48$. Thus

$$E_{j+1} \le \left(\frac{17}{16}K + C_1\right)F_j \le \frac{13}{12}KF_j \le KF_{j+1}.$$

Induction proves the result. No recurrence before J is required. □

Proof of Theorem 3.1: the actual generations. Choose entry constants from Lemma 7.1, and increase the threshold to at least 8. Fix M, S satisfying entry. Apply Corollary 6.4, whose density constant η_{pow} need not equal the entry constant. Lemma 7.1 implies $E_j \le \eta_{\mathrm{pow}}N(m_j)$ eventually. Also $m_j \to \infty$. Thus the complete power recurrence (23) applies to the actual set S_j for every sufficiently large j.

For $X > 0$, the elementary inequality $\log X \le 2X^{\gamma}$ follows from $\log(X^{\gamma}) \le X^{\gamma} - 1$. Hence eventually

$$E_{j+1} \le \frac{17}{16}E_j + (C + 2D)m_j^{\gamma}.$$

Set $F_j = m_j^{\gamma}$. Lemmas 7.2 and 7.3 prove (4). This step keeps the seed fixed while the generation grows; a different seed is not selected at each scale. □

Proof of Theorem 3.1: arbitrary cutoffs. Fix K, J from (4). Given $X > m_J$, choose the first j such that $X \le m_j$. Monotonicity gives $j \ge J$, while minimality gives $m_{j-1} < X$ and

$$m_j = \lfloor 5m_{j-1}/4 \rfloor \le 5X/4.$$

Every permanently missing prime at most X belongs to $\mathcal{P}(m_j) \setminus S_j$. Therefore

$$|\mathcal{P}(X) \setminus \mathcal{C}(S)| \le E_j \le K m_j^\gamma \le K(5/4)^\gamma X^\gamma.$$

Take $K' = K(5/4)^\gamma$. This proves (5) for all sufficiently large integer X. □

8 Birth layers, exact cancellation, and what remains open

The previous results begin with a nearly complete prime prefix. For small-seed growth one must show that a significant part of the representation mass produces *distinct new owners*. The development contains exact tools for that question, but not the needed positive lower bound.

Let A, S be finite prime sets with $\mathcal{T}(A) \subseteq S$, and let B be any finite set of prime targets $r \ge 3$ outside S. Then

$$R_A(r) = 0 \qquad (r \in B). \tag{24}$$

Otherwise a representation from A would put r in $\mathcal{T}(A) \subseteq S$. Thus for every real β,

$$\frac{R_S(r) - \beta R_A(r)}{\sigma(t_r)} = \frac{R_S(r)}{\sigma(t_r)} \qquad (r \in B).$$

This is exact birth-layer cancellation, not an independence heuristic.

If $S \subseteq \mathcal{P}(M)$ and $M \ge 5^8$, the two-prime upper sieve gives

$$0 \le w_r := R_S(r)/\sigma(t_r) \le H(M) := C_2 M/\log^2 M.$$

Writing $W_B = \sum_{r \in B} w_r$, one obtains

$$\sum_{r \in B} \left(\frac{R_S(r) - \beta R_A(r)}{\sigma(t_r)} \right)^2 \le H(M) W_B. \tag{25}$$

Indeed (24) reduces each summand to $w_r^2 \le H(M) w_r$. Since a positive w_r produces r, this also yields

$$W_B \le H(M) |B \cap (\mathcal{T}(S) \setminus S)|. \tag{26}$$

For actual consecutive generations take $A = S_{j-1}$ and $S = S_j$.

These inequalities control the energy in terms of the fresh mass. They do not prove that W_B is positive or grows along the orbit from $\{2\}$. In particular, an upper bound such as (25) cannot be substituted for the lower bound needed in (26). The open small-seed problem is to obtain a usable joint newness/collision estimate that supplies enough fresh mass to reach a certified density region.

8.1 The role of the prime 2

For prime inputs, $2(p+q)-1$ is odd and at least 7, so its largest prime factor is odd. Thus 2 is never created and must be included if the intended closure is to contain every prime. Monotonicity of $\mathcal{T}$ gives the seed-inclusion reductions used in the implementation. The diagonal from p produces $P^+(4p-1)$; it is available at every generation. For instance, $\{3\}$ is not closed: the first diagonal produces 11, then the diagonal from 11 produces 43.

9 Finite verification and quantitative scope

The repository contains a kernel-checked finite certificate

$$\mathcal{P}(10000) \subseteq S_9 \qquad \text{for } S_0 = \{2\}.$$

Its public owner is `inputPrimePrefix_10000_subset_fullOrbit_two_9`. This proves the displayed finite inclusion. It does not bridge the entry threshold in Theorem 3.1.

There is also a separate explicit density-preservation theorem. Writing η_{cert} for the installed certified fraction, one has $\eta_{\mathrm{cert}} \geq 2^{-600}$, and

$$M \geq 2^{49152}, \quad |\mathcal{P}(M) \setminus S| \leq \eta_{\mathrm{cert}} \frac{M}{\log M} \implies |\mathcal{P}(\lfloor 7M/6 \rfloor) \setminus \mathcal{T}(S)| \leq \eta_{\mathrm{cert}} \frac{\lfloor 7M/6 \rfloor}{\log \lfloor 7M/6 \rfloor}.$$

This statement uses expansion factor $7/6$, not $5/4$. It provides a conservative explicit preservation range, not a numerical value for the eventual threshold of the new $79/80$ theorem. The source cutoff 2^{44}, the preservation cutoff 2^{49152}, and the finite certificate at 10000 have different consumers.

The strongest completed dynamical conclusion is therefore the power-saving theorem for sufficiently dense finite seeds. A precise next target is to prove that some actual generation from $\{2\}$ satisfies a fully certified entry condition, or to derive a growth-and-density bootstrap that establishes such entry. Neither a finite diagnostic nor the density result for all prime-pair inputs settles that target. Dense entry is an intermediate milestone toward Conjectures 1.1 and 1.2; it would not by itself remove every exceptional prime.

9.1 Affine and higher-arity generation questions

The broader proposed programme concerns fixed integers

$$a \geq 1, \qquad b \in \mathbb{Z}, \qquad m \geq 2.$$

For a nonempty finite prime seed S, define

$$\mathcal{T}_{a,b,m}(S) = S \cup \{P^+(|a(p_1 + \cdots + p_m) + b|) : (p_1, \ldots, p_m) \in S^m, \\ |a(p_1 + \cdots + p_m) + b| > 1\}.$$

Let $\mathcal{C}_{a,b,m}(S)$ be its increasing iterated closure. Tuples may repeat inputs; permutations and repeated representations do not create multiple copies of an output prime. The absolute value and the exclusion of $0, \pm 1$ specify the convention when b is negative. The present operator is the case $(a, b, m) = (2, -1, 2)$.

We conjecture that complete prime-generation phenomena extend to suitable affine parameters and to sums of more than two prime inputs, with appropriate restrictions on the seed and target primes. A precise classification of those parameters and restrictions remains open. This is a proposed extension, not a theorem covering all triples.

Some necessary restrictions are immediate. If a prime $\ell \mid a$ does not divide b, then every pre-factorization value is nonzero modulo ℓ. Consequently ℓ can belong to the closure only if it is already in the seed. For example, $a = 2, b = -1$ excludes the creation of 2 at every arity. When all available inputs are odd, the value before factorization has parity $am + b$; this must also be retained when formulating a candidate generation statement. A fixed divisor, on the other hand, is not necessarily the largest prime factor, so divisibility alone does not identify the output support.

Congruence restrictions do not exhaust the possible degeneracies. For $(a, b, m) = (1, 0, 2)$, every prime singleton is closed, since $P^+(p+p) = p$. Thus one cannot simply assert universal

generation for every affine map after mentioning modulus caveats. The general problem is to determine which non-degenerate maps and seeds generate every target prime compatible with their arithmetic restrictions, and then prove individual reachability for those cases.

The canonical-owner reduction has a direct higher-arity form. For a target prime r, it is sufficient to find $h \geq 1$, all of whose prime factors are at most r, and available inputs satisfying

$$a(p_1 + \cdots + p_m) + b = hr.$$

Then the largest prime factor is exactly r. If all inputs are odd, the sum must be congruent to m modulo 2. In the subfamily $(a, b) = (2, -1)$, this requires

$$hr \equiv 2m - 1 \pmod 4.$$

For every odd prime r, one of $h = 1$ and $h = 3$ satisfies this congruence, and its prime factors are at most r. This removes the elementary parity obstruction to that owner representation; it does not establish the required additive representation from a generated state. The higher-arity direction therefore preserves the distinction between an arithmetic compatibility check and an actual reachability proof.

A Formal statements and verification map

The maintained proof uses Windows Lean 4.34.0, the repository's pinned Mathlib environment, and released dependency artifacts. Dependencies are reused from their existing caches. The main theorem has no RH, Goldbach, mixing, or small-seed-entry hypothesis. Its substantive seed hypothesis is precisely (3). The analytic facts used to discharge supply and overlap belong to the transitive proof dependency closure.

The following names are relative to `PrimeFactorOscillations.LPF`, except for the final public consumer. Long implementation paths are listed by module stem to keep the table readable.

Mathematical item	Maintained owner or declaration
Full dynamics with diagonals	`LPFDynamics`: `fullStep`, `fullOrbit`. The older `step` and `orbit` definitions exclude diagonals and are not the dynamics of this paper.
Canonical ownership	`LPFJointTransfer`: `primeTarget_lift`, `primeTarget_owner`, `primeTarget_injective_window`.
Uniform character prefix	`LPFCharacterPrefix`: namespace `QRH`, `exists_all_characterChebyshevSum_power_bound`.
Four-packet source identity and budget	`LPFFixedSourceBudget`: `lpf_fixed_source_four_packet_budget`; `LPFSourcePower`: `lpf_fixed_source_power_bound`.
Positive representation supply	`LPFWideSource`: `exists_wide_primeTarget_supply_power_explicit`.
Subset-uniform overlap	`LPFPrimeTargetOverlap`: `exists_uniform_primeTarget_overlap`.
Row normalization and deletion	`LPFRowNormalization`: `primeTargetWeight_self_exact`, `exists_dense_fullStep_prefix_row_normalized_recurrence`.
Complete power/log recurrence	`LPFPositiveSupply`: `exists_fullStep_dense_power_log_bound_from_wide_source`.
Vanishing relative defect	`LPFDenseDensity`: `exists_fullOrbit_dense_density_zero`, with its representation premise discharged by `exists_wide_primeTarget_supply_log`.

Mathematical item	Maintained owner or declaration
Actual-generation 79/80 theorem	`LPFDensePower`: `exists_fullOrbit_dense_power_bound`.
All-cutoff closure theorem	`LPFDensePower`: `fullOrbitMissingPrimes`, `exists_fullOrbit_dense_closure_power_bound`.
Public implementation consumer	`Solution.lean`: `PrimeFactorOscillations.publicLPFDenseClosurePowerBound`.
Fresh-owner energy and mass	`LPFWeightedCanonicalEnergy`: `fresh_birth_energy_upper`, `fresh_mass_bound`.
Finite seed certificate	`LPFFiniteEntry.Result`: `inputPrimePrefix_10000_subset_fullOrbit_two_9`.
Separate explicit preservation	`LPFNumericalPreservation`: `lpf_density_preservation_numerical_cutoff`.

The closure statement is formalized over integer cutoffs. In readable quantifier form it says

$$\begin{aligned}&\exists \eta > 0 \ \ \forall M \text{ sufficiently large } \ \forall S,\\ &\quad |\mathcal{P}(M) \setminus S| \le \eta M/\log M\\ &\qquad \Longrightarrow \ \exists K > 0 \ \ \forall X \text{ sufficiently large},\\ &\qquad\qquad |\mathcal{P}(X) \setminus \mathcal{C}(S)| \le K X^{79/80}.\end{aligned}$$

The implementation represents the final missing set by filtering the finite prime prefix with the condition that membership fails at every generation. No infinite-set cardinality is used.

The formal proof of the new power-saving conclusion proceeds in the same order as Section 7: cutoff power growth, an eventual recurrence lemma, the transfer from vanishing density to the power recurrence, and first-crossing interpolation. These are proofs, not axioms encoding the conclusion. The declaration audit checks their complete axiom sets; standard foundational axioms such as propositional extensionality, classical choice, and quotient soundness are distinguished from any additional analytic assumption. The build and audit commands, artifact pins, and saved results are supplied in the companion repository.

B What the result does and does not transfer

The result concerns an actual iterated closure once finite entry is given. It is stronger than a limiting divisibility law for the full prime-pair ensemble: such a law alone does not describe a dynamically generated input set. Conversely, it is weaker than all-prime reachability. A power-saving exceptional set may still contain infinitely many primes.

No criterion for the Riemann hypothesis or for bounded prime gaps is derived from this dynamical theorem. The companion paper on prime-factor densities treats those separate questions with different quantifiers and different signed inequalities. Sharing analytic tools does not supply an implication between the two sets of conclusions.

References

[1] G. Back and M. Caragiu, *The greatest prime factor and recurrent sequences*, Fibonacci Quarterly **48** (2010), no. 4, 358–362. https://fq.math.ca/Papers1/48-4/Back_Caragiu.pdf.

[2] M. Caragiu and P. A. Vicol, *Prime magmas and a cyclicity conjecture*, JP Journal of Algebra, Number Theory and Applications **38** (2016), no. 2, 129–143. doi:10.17654/NT038020129.

[3] O. Ramaré and I. Z. Ruzsa, *Additive properties of dense subsets of sifted sequences*, Journal de Théorie des Nombres de Bordeaux **13** (2001), no. 2, 559–581. doi:10.5802/jtnb.338.

[4] B. Green and T. Tao, *Restriction theory of the Selberg sieve, with applications*, Journal de Théorie des Nombres de Bordeaux **18** (2006), no. 1, 147–182. https://jtnb.centre-mersenne.org/article/JTNB_2006__18_1_147_0.pdf.

[5] A. Alsetri and X. Shao, *Density versions of the binary Goldbach problem*, Acta Arithmetica **218** (2025), no. 3, 285–295. Preprint arXiv:2405.18576v2, 19 September 2024. https://arxiv.org/abs/2405.18576v2.

[6] OpenAI, *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane* $\mathrm{Re}(s) > 7/8$, preprint, 30 September 2026, with companion Lean development. https://github.com/openai/math/tree/main/preprints/The-Quasi-Riemann-Hypothesis-September-30-2026. The referenced nonvanishing source is also available at revision adc7f1241b42e322a6451854ab7e4b4c146bf78a of that repository.

[7] J. Whidden, *Prime Factor Oscillations*, companion source and formal proofs, 2026. https://github.com/kimihiro64/prime-factor-oscillations.